
Online Examination Report

Date:	28.07.2014
Attempts:	1
Examinee:	Atwoki Kaija, Albert
Date of birth:	
Subject:	080- Principles of flight
Result:	0 %
Time used:	00:00:15 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	205	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
2	221	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
3	197	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
4	208	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
5	213	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
6	185	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
7	240	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
8	217	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
9	184	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
10	206	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
11	189	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
12	190	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
13	210	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
14	229	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
15	211	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
16	215	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
17	58	81.1.1.1	Laws and definitions	0,00	1,00
18	61	81.1.1.4	Shape of an aerofoil section	0,00	1,00
19	30	81.1.2	Two-dimensional airflow around an aerofoil	0,00	1,00
20	100	81.1.3.1	The lift coefficient C_l	0,00	1,00
21	47	81.1.4.1	Streamline pattern	0,00	1,00
22	93	81.1.4.2	Induced drag	0,00	1,00
23	4	81.1.4.2	Induced drag	0,00	1,00
24	70	81.1.4.2	Induced drag	0,00	1,00
25	16	81.2.2.2	Oblique shock waves	0,00	1,00
26	59	81.2.2.2	Oblique shock waves	0,00	1,00
27	104	81.4	STABILITY	0,00	1,00
28	101	81.4	STABILITY	0,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	14	81.4.1.1	Basics and definitions	0,00	1,00
30	87	81.4.3.3	Neutral point	0,00	1,00
31	99	81.4.6.3	Dutch roll	0,00	1,00
32	10	81.5.8.3	Stabiliser trim	0,00	1,00
33	98	81.5.8.3	Stabiliser trim	0,00	1,00
34	96	81.6.1.1	Flutter	0,00	1,00
35	109	81.7	PROPELLERS	0,00	1,00
36	1	81.7.4.4	Asymmetric blade effect	0,00	1,00
37	69	81.8.1.2	Straight steady climb	0,00	1,00
38	65	81.8.1.2	Straight steady climb	0,00	1,00
39	68	81.8.1.5	Steady coordinated turn	0,00	1,00
40	3	81.8.1.5	Steady coordinated turn	0,00	1,00
Total:0%				0,00	40,00

Attachment 2: List of result by topic

Licence

Date: 28.07.2014

Examinee: Atwoki Kaija, Albert

Location: Uganda Civil Aviation Authority

Your Result: 0,00 from 40,00 Points (0,00 %)

Topics	Amount of Questions	Points achieved	Maximum Points	Percent
81. PRINCIPLES OF FLIGHT AEROPLANE	40	0,00	40,00	0,00
81.1. SUBSONIC AERODYNAMICS	8	0,00	8,00	0,00
81.2. HIGH SPEED AERODYNAMICS	2	0,00	2,00	0,00
81.4. STABILITY	5	0,00	5,00	0,00
81.5. CONTROL	2	0,00	2,00	0,00
81.6. LIMITATIONS	1	0,00	1,00	0,00
81.7. PROPELLERS	2	0,00	2,00	0,00
81.8. FLIGHT MECHANICS	4	0,00	4,00	0,00
Total	40	0,00	40,00	0,00

1

What could result if a glider pilot releases while in the low-tow position during an aerotow?

- ☐ Nose of the glider would tend to pitch up after release.
- ☐ Glider may be forced into the towplane's wake turbulence.
- ☒ Tow ring may strike and damage the glider after release.

2

Dampers should normally be kept closed during a climb to altitude because any air blown into the system would

- ☒ **increase the amount of air to be valved, resulting in a slower rate of ascent.**
- ☐ **decrease the volume of gas within the envelope.**
- ☐ **increase the amount of gas to be valved, preventing the airship from ascending too fast.**

3

If the gyroplane's CG is below the propeller thrust line, which direction will the application of power cause the nose to move?

- ☒ The nose will pitch down.
- ☐ The nose will not move.
- ☐ The nose will pitch up.

4

During a winch launch, which factor would most likely result in pitch oscillations?

- ☐ Winching speed too fast.
- ☒ Insufficient up-elevator control.
- ☐ Winching speed too slow.

5

Which is true relating to the direction in which turns should be made during slope soaring?

- ☐ All reversing turns should be made to the left.
- ☒ All reversing turns should be made into the wind away from the slope.
- ☐ All turns should be made downwind toward the slope.

6

If the landing gear on an airplane moves forward during retraction, the total moment will

☒ decrease.

☐ increase.

☐ remain the same.

7

(Refer to Figure 146.) Identify the system that has failed and determine a corrective action to return the airplane to straight-and-level flight.

- ☐ Electrical system has failed; reduce power, roll left to level wings, and raise the nose to reduce airspeed.
- ☐ Vacuum system has failed; reduce power, roll left to level wings, and pitchup to reduce airspeed.
- ☒ Static/pitot system is blocked; lower the nose and level the wings to level-flight attitude by use of attitude indicator.

See Attachments:

1. instrument_146.jpg

figure 146

(Attachment
No.1)

8

What is the relationship of false lift to the wind? False lift

- ☐ increases if the vertical velocity of a balloon increases.
- ☒ decreases as the wind accelerates a balloon to the same speed as the wind.
- ☐ exists only if the surface winds are calm.

9

If the nosewheel of an airplane moves aft during gear retraction, how would this aft movement affect the CG location of that airplane? It would

- ☒ **cause the CG location to move aft.**
- ☐ **cause the CG location to move forward.**
- ☐ **have no effect on the CG location.**

10

During an autolaunch, the pitch angle of the glider should not exceed

- ☐ 15 degrees at 50 feet, 20 degrees at 100 feet, and 40 degrees at 200 feet.
- ☐ 10 degrees at 50 feet, 20 degrees at 100 feet, and 45 degrees at 200 feet.
- ☒ 15 degrees at 50 feet, 30 degrees at 100 feet, and 45 degrees at 200 feet.

11

When one engine fails on a twin-engine airplane, the resulting performance loss

- ☒ may reduce the rate of climb by 80 percent or more.
- ☐ is approximately 50 percent since 50 percent of the normally available thrust is lost.
- ☐ reduces cruise indicated airspeed by 50 percent or more.

12

Maximum gliding distance of an aircraft is obtained when

- ☒ induced drag and parasite drag are equal.
- ☐ parasite drag is the least.
- ☐ induced drag equals the coefficient of lift.

13

A rule of thumb for flying a final approach is to maintain a speed that is

- ☐ 50 percent above the glider's stall speed, regardless of windspeed.
- ☒ 50 percent above the glider's stall speed plus half the estimated windspeed.
- ☐ twice the glider's stall speed plus half the estimated windspeed.

14

How does the wing design feature 'washout' affect the production of lift?

- ☒ The wing tips continue producing lift when the main body of the wing is not producing lift.
- ☐ The center of lift moves from the trailing edge of the wing, to the leading edge of the wing, as the wing begins to stall.
- ☐ The main body of the wing continues to produce lift when the wing tips are not producing lift.

15

If swirling dust, leaves, or debris indicate a strong thermal on the final approach to a landing, it is recommended that the glider pilot

- ☒ **close the spoilers and increase the airspeed.**
- ☐ **open the spoilers and reduce the airspeed.**
- ☐ **open the spoilers and maintain a constant airspeed.**

16

Which would most likely ensure a safe off-field landing?

- ☐ Landing into the wind, regardless of the type or slope of the terrain.
- ☒ Maintaining an approach airspeed of at least 50 percent above the glider's stall speed plus half the estimated windspeed.
- ☐ Landing in a pasture or uncultivated field rather than one in cultivation and whose crops have been harvested.

17

Considering subsonic incompressible airflow through a Venturi, which statement is correct?

- I. The dynamic pressure in the throat is lower than in the undisturbed airflow.
II. The total pressure in the throat is the same as in the undisturbed airflow.

- ☐ I is correct, II is correct.
- ☒ I is incorrect, II is correct.
- ☐ I is incorrect, II is incorrect.
- ☐ I is correct, II is incorrect.

18

An aerofoil is cambered when:

- ☒ the line, which connects the centres of all inscribed circles, is curved.
- ☐ the upper surface of the aerofoil is curved.
- ☐ the maximum thickness is large compared with the length of the chord.
- ☐ the chord line is curved.

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Subject: 080- Principles of flight
Date: 2014-07-28

19

(Please view annex 081-0005 issue date July 2004)

How are the speeds at point 1 and point 2 related in the figure to the relative wind/airflow V?

☐ $V_1 = 0$ and $V_2 = V$.

☐ $V_1 < V_2$ and $V_2 < V$.

☒ $V_1 = 0$ and $V_2 > V$.

☐ $V_1 > V_2$ and $V_2 < V$.

See Attachments:

1. 081-0005.gif

Annex

(Attachment
No.2)

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Subject: 080- Principles of flight
Date: 2014-07-28

20

(Please view annex 081-0006 issue date July 2004)

The point in the annex corresponding to CL for minimum horizontal flight speed is:

☐ point b.

☐ point a.

☒ point c.

☐ point d.

See Attachments:

1. 081-0006.gif

Annex

(Attachment
No.3)

21

Which of these statements about the strength of wing tip vortices are correct or incorrect?

- I. Assuming no flow separation, the strength of wing tip vortices decreases as the angle of attack decreases.
- II. The strength of wing tip vortices increases as the aspect ratio decreases.

- ☐ I is correct, II is incorrect.
- ☐ I is incorrect, II is incorrect.
- ☐ I is incorrect, II is correct.
- ☒ I is correct, II is correct.

22

Which of these statements about induced drag are correct or incorrect?

- I. An elliptical spanwise lift distribution generates more induced drag than a rectangular lift distribution.
II. Induced drag decreases with decreasing aspect ratio.

- ☐ I is correct, II is incorrect.
- ☒ I is incorrect, II is incorrect.
- ☐ I is incorrect, II is correct.
- ☐ I is correct, II is correct.

23

Which statement concerning the local flow pattern around a wing is correct?

- ☐ Slat extension, at a constant angle of attack and normal extension speeds, will increase the lift coefficient,
- ☐ Vortex generators on the wing partially block the spanwise flow over the wing
- ☒ By fitting winglets to the wing tip, reduces induced drag.
- ☐ Sweepback reduces drag since, because the span increases.

24

Which of the following wing shapes produces the lowest induced drag? (assume zero wing twist)

☐ Tapered.

☐ Rectangular.

☒ Elliptical.

☐ Circular.

25

Compared with an oblique shock wave a normal shock wave has a:

- ☒ higher loss in total pressure.
- ☐ higher total temperature.
- ☐ lower static temperature.
- ☐ higher total pressure.

26

Which of these statements about an oblique shock wave are correct or incorrect?

- I. The static temperature behind an oblique shock wave is lower than in front of it.
II. The static pressure behind an oblique shock wave is higher than in front of it.

- ☒ I is incorrect, II is correct.
- ☐ I is incorrect, II is incorrect.
- ☐ I is correct, II is correct.
- ☐ I is correct, II is incorrect.

27

If an aircraft has negative dynamic and positive static stability, this will result in

- ☐ undamped oscillations.
- ☐ convergent oscillations.
- ☒ divergent oscillations.

28

If the aircraft's nose initially tends to move farther from its original position after the elevator control is pressed forward and released, the aircraft displays

- ☐ positive static stability.
- ☐ positive dynamic stability.
- ☒ negative static stability.

29

A statically unstable aeroplane:

- ☐ is always dynamically stable.
- ☐ reduces its instability as the CG moves aft.
- ☒ can never be dynamically stable.
- ☐ will become stable at high speed.

30

Which of the following statements about static longitudinal stability is correct?

- I. A requirement for positive static longitudinal stability of an aeroplane is, that the neutral point is behind the centre of gravity.
- II. A wing with positive camber provides a positive contribution to static longitudinal stability, when the centre of gravity of the aeroplane is in front of the aerodynamic centre of the wing.

- ☐ I is incorrect, II is correct.
- ☐ I is incorrect, II is incorrect.
- ☐ I is correct, II is incorrect.
- ☒ I is correct, II is correct.

31

An aeroplane is sensitive to Dutch roll when:

- ☐ static stability is less pronounced than dynamic stability.
- ☐ static lateral stability is much less pronounced than static directional stability.
- ☒ static lateral stability is much more pronounced than static directional stability.
- ☐ an aeroplane has anhedral.

32

Comparing the differences between a horizontal trimmable stabiliser and an elevator trim tab, which statement is correct?

- I. A stabiliser trim is less suitable for jet transport aeroplanes because of their large speed range.
II. A trim tab is a more powerful means of trimming.

☒ I is incorrect, II is incorrect.

☐ I is correct, II is correct.

☐ I is incorrect, II is correct.

☐ I is correct, II is incorrect.

33

What is the effect on landing speed when a trimmable horizontal stabiliser jams at high IAS?

- ☐ In most cases, no effect.
- ☒ In most cases, a higher than normal landing speed is required.
- ☐ No effect with a forward CG.
- ☐ No effect when landing on a high elevation runway.

34

Wing flutter may be caused by a:

- ☒ combination of bending and torsion of the wing structure.
- ☐ combination roll control reversal and low speed stall.
- ☐ aerodynamic wing stall at high speed.
- ☐ combination of fuselage bending and wing torsion.

35

A propeller rotating clockwise, as seen from the rear, creates a spiraling slipstream that tends to rotate the aircraft to the

- ☐ right around the vertical axis, and to the left around the longitudinal axis.
- ☐ left around the vertical axis, and to the left around the longitudinal axis.
- ☒ left around the vertical axis, and to the right around the longitudinal axis.

36

Which statement about a propeller is correct?

- I. Asymmetric blade effect is unaffected when engine power is increased.
II. Asymmetric blade effect is independent of the angle between the propeller axis and the airflow through the propeller disc.

- ☒ I is incorrect, II is incorrect.
- ☐ I is correct, II is correct.
- ☐ I is correct, II is incorrect.
- ☐ I is incorrect, II is correct.

37

Given:
Aeroplane mass: 50 000kg.
Lift/Drag ratio: 12.
Thrust per engine: 28 000N.
Assumed g: 10m/s².

For a straight, steady, wings level climb of a three-engine aeroplane, the one-engine inoperative climb gradient is:

☐ 9.7%.

☐ 8.5%.

☒ 2.9%.

☐ 8.0%.

38

Given:
Aeroplane mass: 50 000kg.
Lift/Drag ratio: 10.
Thrust per engine: 30 000N.
Assumed g: 10m/s².

For a straight, steady, wings level climb of a three-engine aeroplane, the all-engines climb gradient is:

☐ 2.9%.

☐ 9.7%.

☒ 8.0%.

☐ 8.5%.

39

An aeroplane performs a steady horizontal, co-ordinated turn with 45 degrees of bank at 230 kt TAS. The same aeroplane with the same bank angle and speed, but at a higher mass:

- ☐ will have a higher rate of turn.
- ☒ will turn with the same radius, but might stall.
- ☐ will turn with a smaller turn radius.
- ☐ will turn with a larger turn radius.

40

An aeroplane performs a steady horizontal, co-ordinated turn with 45 degrees of bank at 230 kt TAS. The same aeroplane with the same bank angle and speed, but at a lower mass will turn with:

- ☐ a lower rate of turn.
- ☒ the same rate of turn.
- ☐ a smaller turn radius.
- ☐ a larger turn radius.

Attachment No. 1

Questions 7

figure 146

instrument_146.jpg

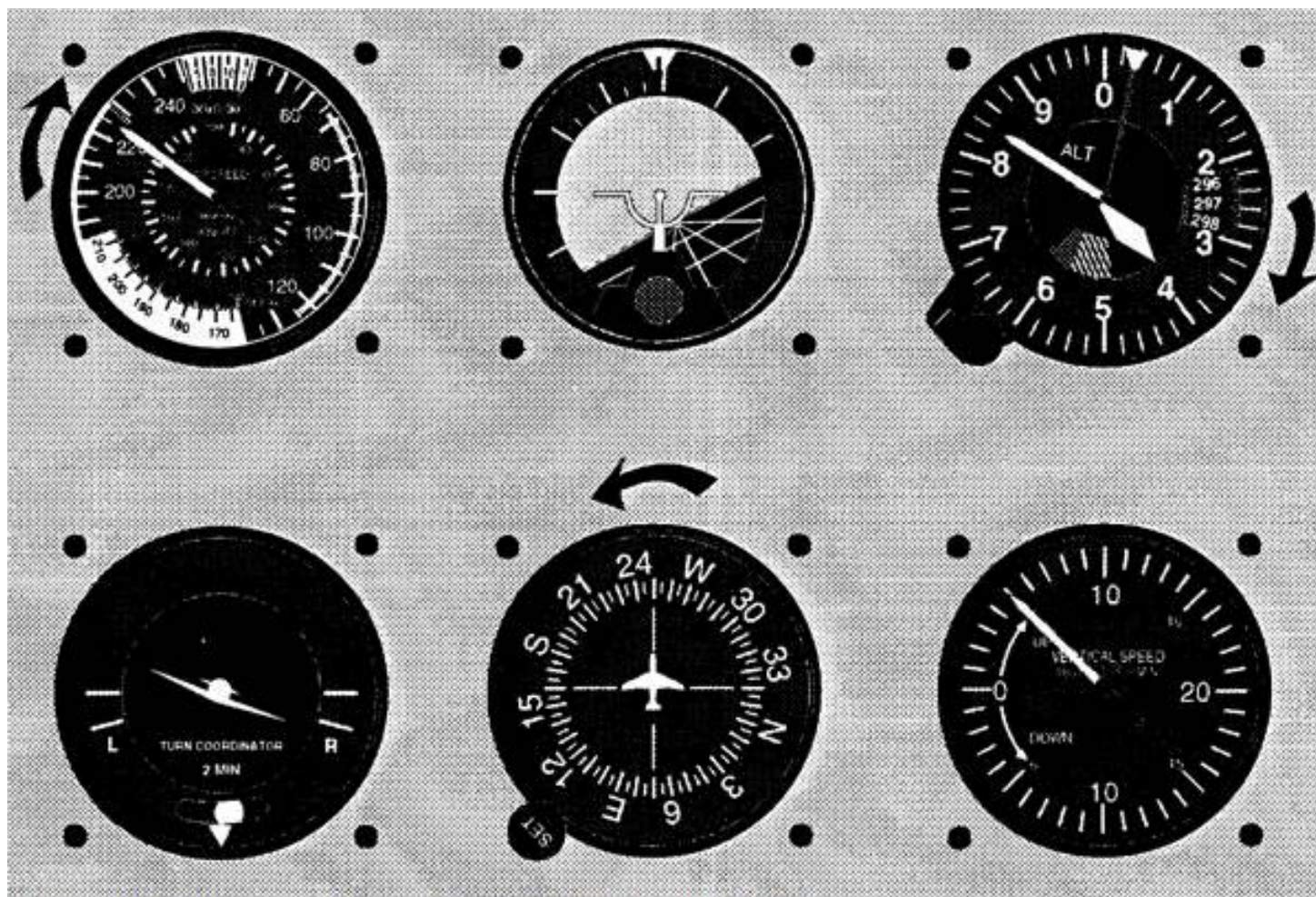


Figure 146. Instrument Sequence (System Failed) © ASA

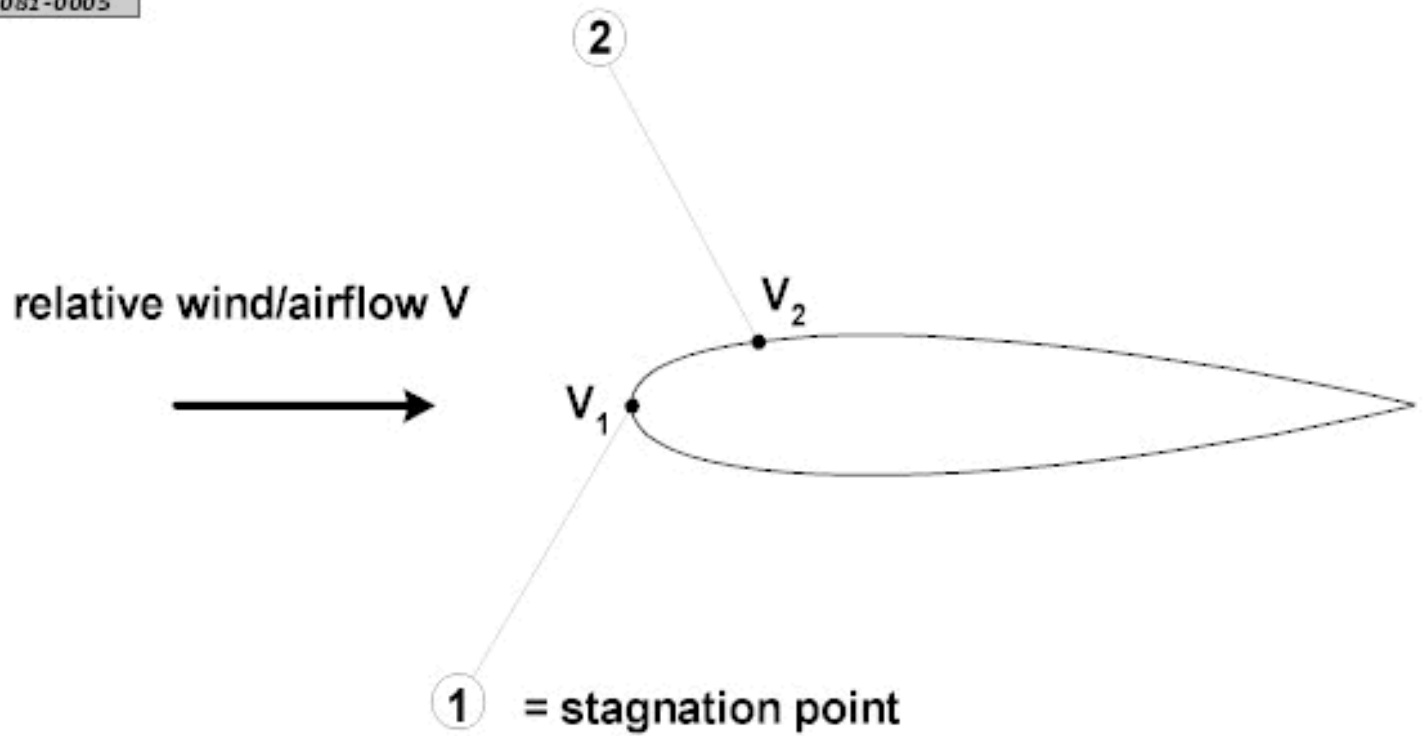
Attachment No. 2

Questions 19

Annex

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Attachment No. 3

Questions 20

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